

# FIPS 140-3 (Cryptographic Module Validation)

**\*\*Anticloud FZ LLE — PAX L5 Narrow L2 General 27B\*\***  
Run: 2026-10-02T09:35:10.925858Z | Framework ID: 14\_FIPS\_140\_3

## Standard

FIPS 140-3 (NIST, 2019), ISO/IEC 19790:2012

## Result

PARTIAL — SHA3-256 (FIPS-approved), AES-256-GCM (FIPS-approved); script key derivation not FIPS-approved

## Score

Level 2 target — SHA3/AES approved; script requires substitution with PBKDF2/HKDF for full FIPS

## Notes

SHA3-256 (FIPS 202), AES-256-GCM (SP 800-38D) are FIPS-approved. Recommendation: replace script with PBKDF2-SHA512.

## Real Benchmark Numbers (Kaggle T4, loiskleinner/pax-benchmark-runner v3)

| Module                  | Key Metric           | Value       |
|-------------------------|----------------------|-------------|
| CRDT                    | lww_set_us           | 0.51        |
| CRDT                    | merge_10k_ms         | 0.07        |
| CRDT                    | convergence          | GUARANTEED  |
| CryptographicProvenance | write_per_receipt_ms | 0.059       |
| CryptographicProvenance | verify_1k_chain_ms   | 2.53        |
| ZeroTrustLocalMesh      | ca_gen_ms            | 367.5       |
| ZeroTrustLocalMesh      | cert_issue_ms        | 333.8       |
| ZeroTrustLocalMesh      | tls_version          | 1.3         |
| LocalObservabilityStack | spans_per_sec        | 17558       |
| LocalObservabilityStack | logs_per_sec         | 24492       |
| SovereignMemory         | write_per_entry_ms   | 51.94       |
| SovereignMemory         | encryption           | AES-256-GCM |
| AIOSS_AuditChain        | entries_per_sec      | 232665      |

| AIOSS\_AuditChain | hash\_algo | SHA3-256 |

Environment Provenance (Preprioception)

| Field | Value |  
|-----|-----|  
| Run ID | loiskleinner/pax-benchmark-runner v3 |  
| Run Date | 2026-10-02 |  
| Platform | Kaggle T4 x2 GPU, Ubuntu 22.04, Python 3.12 |  
| GPU | NVIDIA Tesla T4 (2× 16 GB VRAM) |  
| CPU | Intel Xeon (2 vCPU, 13 GB RAM) |  
| Inference Temperature | N/A (code benchmarks, no LLM sampling) |  
| KV Cache | N/A (no LLM inference in this cell) |  
| Framework | llama-cpp-python 0.3.x (installed from source with CUDA) |

Cryptographic Integrity

| Field | Value |  
|-----|-----|  
| SHA3-256 (content) | 16055c2d7559282dd803f09a0894518dcc392004198fc069d25930d28cda702d |  
| SHA3-256 (chain) | c535f5793573a48d66f7e90d53ce51d6471f197f4959d355bcf241457393b35b |  
| Timestamp UTC | 2026-10-02T09:35:10.925858Z |  
| Hash Algorithm | SHA3-256 (FIPS 202 compliant) |

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\*Anticloud FZ LLE · PAX L5 Narrow L2 General 27B · AIOSS Ledger Genesis:  
8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560\*